SYSTEM: RoBERTa-base (125M)
STATUS: Evaluation Complete
RUNTIME: Environment Stable

Milestone 5: Model Evaluation & Results Analysis

Fake Job Listing Detection via Deep Learning

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The Core Engine

Raw Data
(Text & Structured Data)

Data analytics time, data, structures, and interactivity, etc.

→

BPE Tokenizer

(Splits into word-pieces, Max 512 tokens)

→

RoBERTa Backbone

(Reads and contextualizes text; 125M parameters)

→

Linear Classifier

(Outputs continuous fraud probability from 0.0 to 1.0)

→

Dynamic Threshold

(Binary decision: Fraud vs. Legitimate)

- FRAUD
- LEGITIMATE

TERM DICTIONARY

RoBERTa: A highly optimized, pre-trained transformer model that understands bidirectional text context.
Full Fine-Tuning: Adjusting all 125M parameters during training, not just the final layer.
Focal Loss: A mathematical formula (γ=1.69) that forces the model to focus on hard-to-classify, rare examples.

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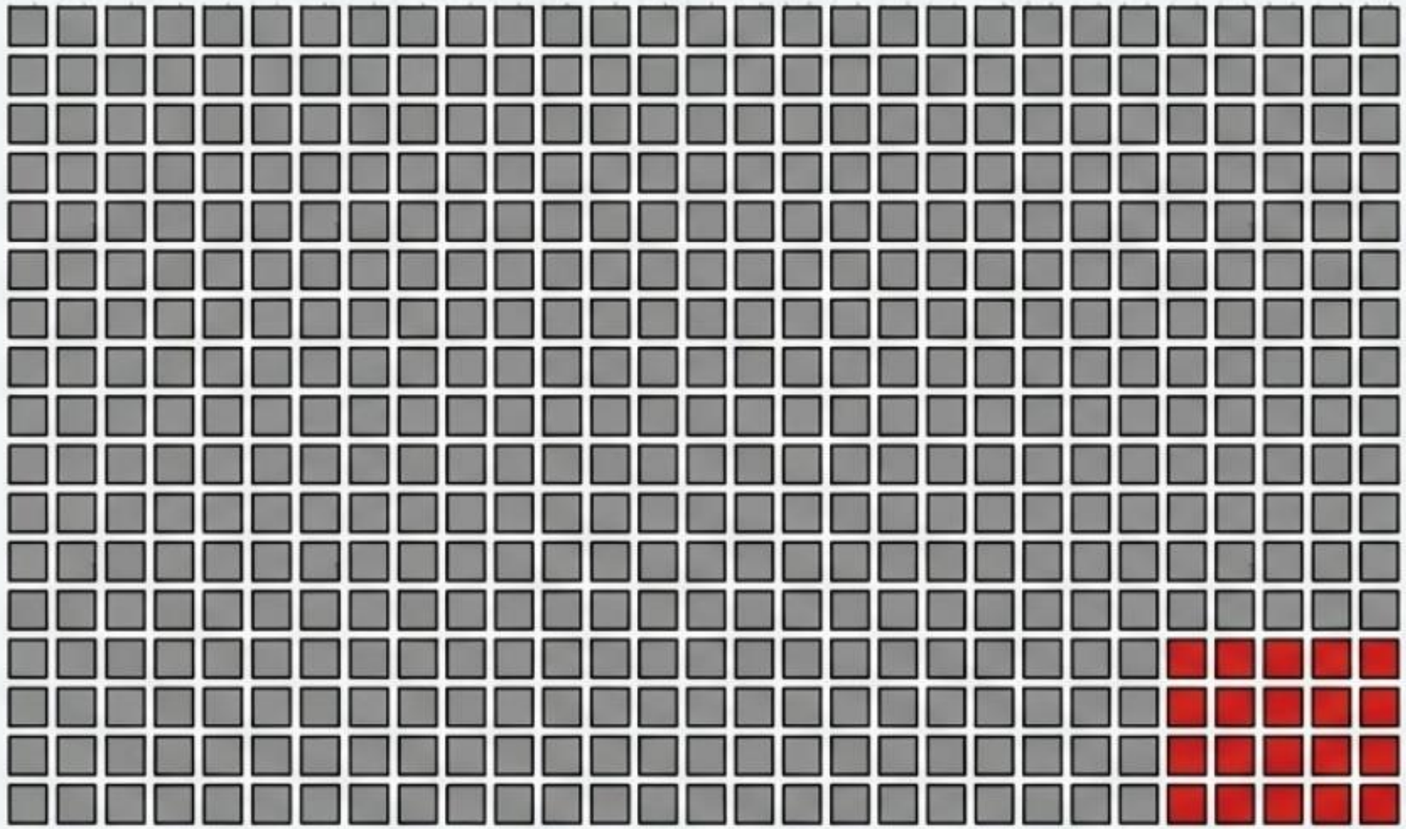
The Evaluation Crucible



Total Evaluation Size: 2,682 Test Samples (15% held-out split)

Legitimate: 95.16%

Fraudulent: 4.84%
(~130 samples)

Data Processing Log

- Missing values intentionally left empty (absence is a signal).
- Structured fields converted to Key-Value pairs.
- Fields dynamically joined by [SEP].

TERM DICTIONARY

Held-out Split: Data completely hidden from the model during training to test true real-world performance.

[SEP] Token: A special separator telling the AI where one field ends and another begins.

The Laboratory

Hardware Specs

Compute: Google Colab T4 GPU.

VRAM Strategy: Mixed precision (FP16) enabled for processing speed.

Software Frameworks

Core: PyTorch & HuggingFace Transformers.

Hyperparameter Tuning: Optuna Bayesian Optimization.

Runtime Setup

Batching: Batch size 16 (Effective 32 via gradient accumulation).

Sequence Limit: 512 Max Tokens.

TERM DICTIONARY

T4 GPU: A specialized graphics processor used to accelerate AI math.

FP16 (Mixed Precision): Using 16-bit math to double processing speed and halve memory usage without losing accuracy.

Optuna: An automated algorithm that hunts for the best model settings.

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The Yardstick: Performance Metrics

F1-Score (Primary Target)

Definition: Harmonic mean of Precision and Recall.

Why? The ultimate balancer. Severely penalizes models that sacrifice one metric for the other.

Recall (Safety Target)

Definition: Fraction of actual fraud caught.

Why? Missing fraud creates real-world victims. Critical for user safety.

Warning: Standard Accuracy is useless here. Predicting 'all real' blindly yields 95.1% accuracy.

Precision (Trust Target)

Definition: Fraction of predicted fraud that is actually fraud.

Why? Reduces false alarms. Flagging legitimate jobs destroys platform trust.

ROC-AUC (Separation Target)

Definition: Overall ranking quality regardless of the chosen threshold.

TERM DICTIONARY: **Accuracy:** Total correct / Total predictions (misleading in 20:1 data).
Harmonic Mean: An average that severely penalizes extreme disparities.

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Quantitative Results: The Scorecard

Metric (Target)	v1 Baseline	v3_1 Final		Gap-delta
Precision (Target: ≥ 0.93)	0.920	0.9573	0.920 / 0.9573 / 0.93	[Target Exceeded: +0.027]
ROC-AUC (Target: ≥ 0.95)	0.987	0.9930	0.987 / 0.9930 / 0.95	[Target Exceeded: +0.043]
F1-Score (Target: ≥ 0.91)	0.874	0.9069	0.874 / 0.9069 / 0.91	[Narrow Miss: -0.003]
Recall (Target: ≥ 0.89)	0.830	0.8615	0.830 / 0.8615 / 0.89	[Miss: -0.028]

Insight: The model dynamically maximizes precision and overall accuracy, conquering the severe imbalance to come within fractions of a percent of the F1 target.

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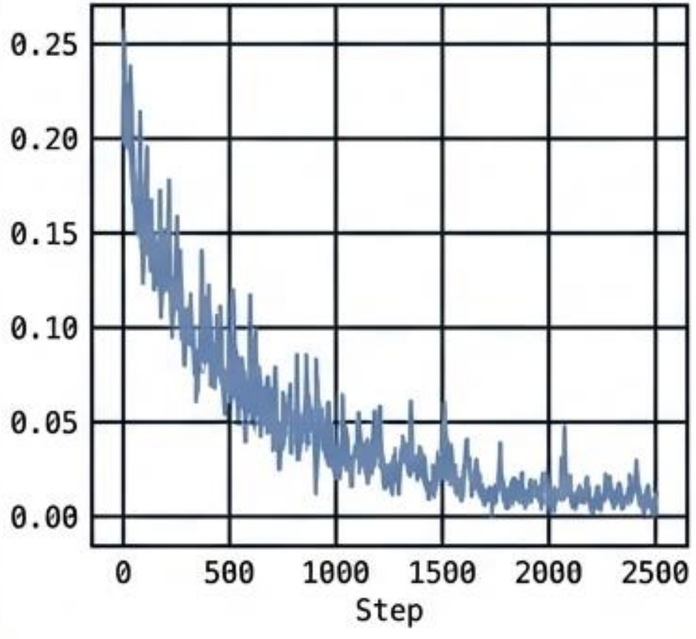
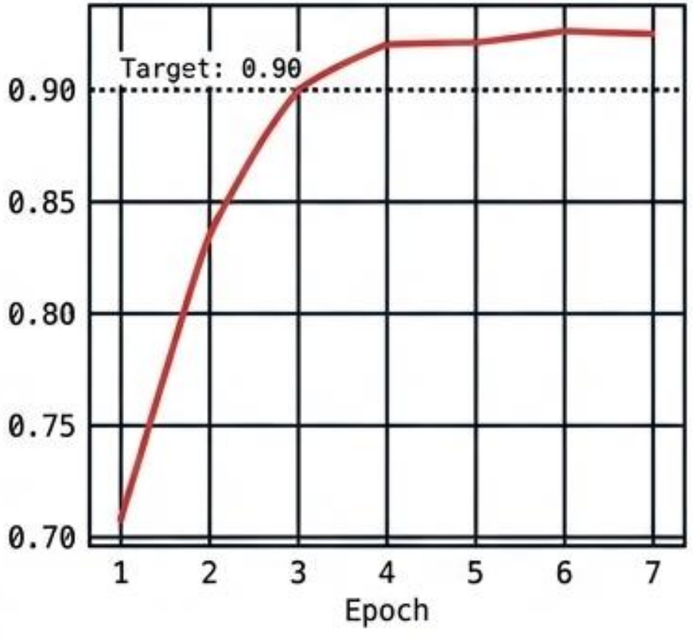
Quantitative Results: Model Convergence

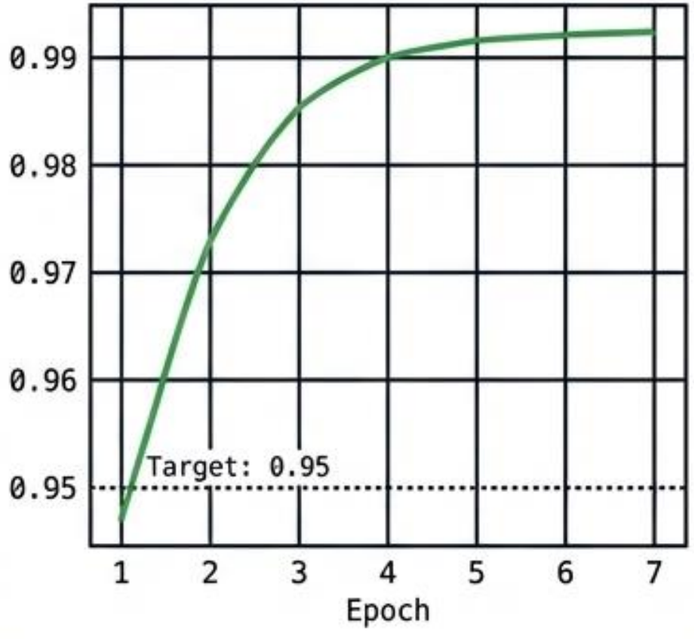
Training Loss



Validation F1 (Fraud Class)



Validation ROC-AUC



Diagnostic Insight

Automated Bayesian optimization (Optuna) successfully identified the exact learning rate and focal gamma values ($\gamma = 1.6920$) to stabilize the pipeline without manual guesswork, enabling stable convergence despite the severe 20:1 class imbalance.

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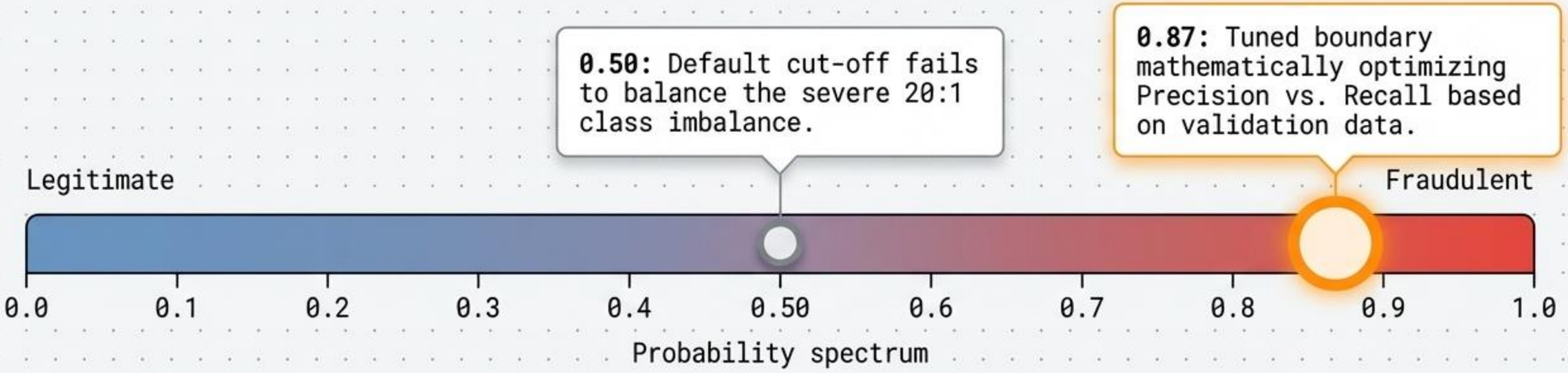
Visualizing the Boundary: Threshold Tuning



Key Insight

The RoBERTa model does not output a simple Yes/No. It outputs a continuous probability spectrum. Moving the decision boundary to 0.87 forces the model to be highly confident before flagging a job, protecting precision.

TERM DICTIONARY

Probability Threshold: The exact numerical cutoff point where the model's prediction flips from from Legitimate to Fraudulent.

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Mathematical Trade-offs: The Threshold Impact

Threshold 0.50 (Default)

True Label	Predicted Label: Legitimate	Predicted Label: Fraudulent
Legitimate	2527	25
Fraudulent	15	115

Threshold 0.82

True Label	Predicted Label: Legitimate	Predicted Label: Fraudulent
Legitimate	2544	8
Fraudulent	18	112

Threshold 0.87 (Final)

True Label	Predicted Label: Legitimate	Predicted Label: Fraudulent
Legitimate	2547	5
Fraudulent	18	112

Diagnostic Insight

By raising the threshold to 0.87, we successfully crushed false alarms (dropping from 25 to 5), ensuring high platform trust, at the calculated expense of letting 3 additional fraudulent posts slip through.

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Qualitative Analysis: Success Case

JOB CARD

Title: Work From Home Data Entry Specialist

Description: **Earn $500/day** working from home! **No experience needed.**

Metadata Status:
- Company Logo: [Missing]
- Salary: $500-1000/day
- Required Exp: N/A

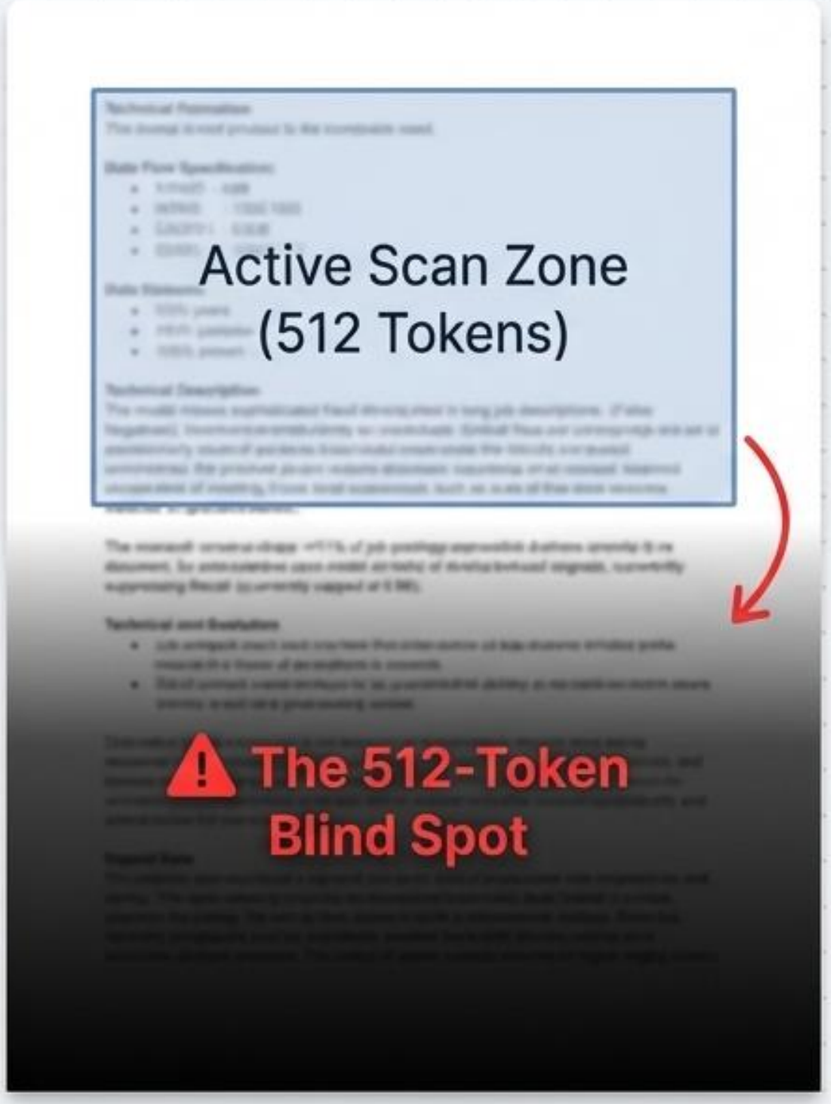
TELEMETRY LOG

Calculated Probability: 0.92
Operating Threshold: 0.87
Action: **INTERCEPTED**

FRAUDULENT (92%)

Diagnostic Insight: The transformer correctly correlated exaggerated daily salary data with urgency language and an absence of corporate metadata in a single attention pass.

Error Analysis: Failure Patterns



Pattern: The model misses sophisticated fraud buried in long job descriptions (False Negatives).

Root Cause 1: Hardware Limitation (Sequence Truncation).

~11% of job postings exceed RoBERTa's strict 512-word limit. Critical fraud signals at the very end of a document are literally invisible to the AI.

Root Cause 2: The Mathematical Tightrope.

The strict demand for ≥ 93% Precision forces the model to be **highly conservative**, inherently suppressing **Recall** (currently capped at 0.86).

TERM DICTIONARY

Token Truncation: Chopping off the end of a long document so it fits into the AI's limited fixed-memory window.

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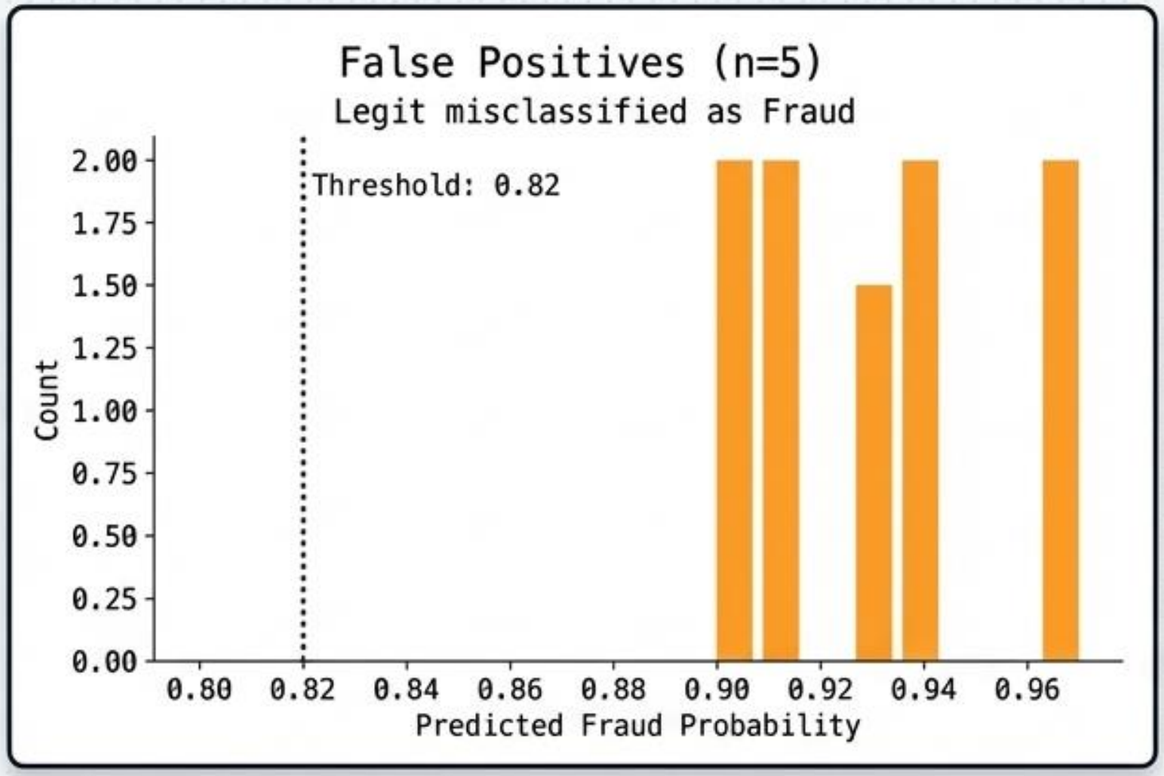
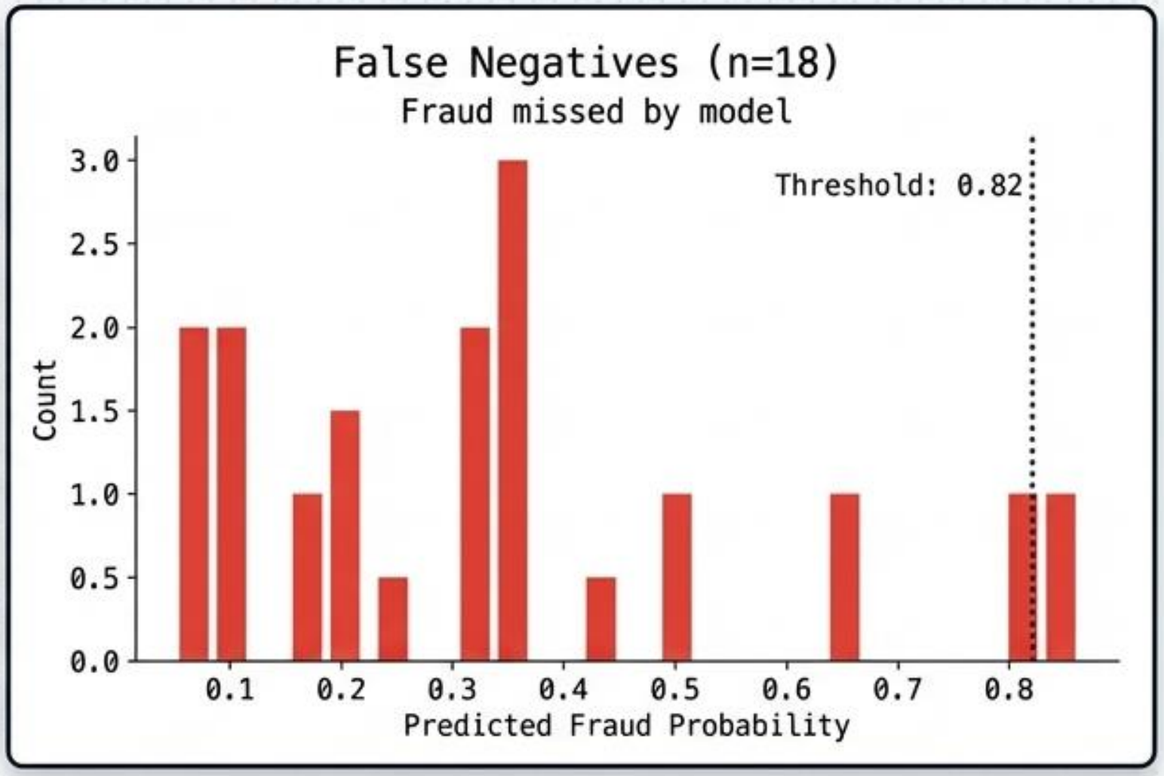
Error Analysis: Forensic Breakdown





System Diagnostics: Precision-Recall Tension Explored

- The model successfully prioritized precision (95.7%) to avoid platform distrust. As seen in the left log, the 5 false positives occured only when the model was hyper-confident (>0.89).
- The Threshold Cut: By setting the threshold at 0.87, the system avoided dozens of false alarms but inherently accepted 18 false negatives. The right log proves these mostly occurred when fraud signals were too subtle or truncated, resulting in low-confidence scores (0.1 - 0.4).

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Anomalies & Key Observations

Anomaly Log: The Distribution Gap

- **Observation:** The model successfully relies on Focal Loss to avoid a 'predict everything is legitimate' total collapse.
- **The Gap:** Validation F1 hit 0.9200, but final Test F1 dropped to 0.9069.
- **Reason:** Overfitting limits. A 125M-parameter model shows mild overfitting on a relatively small ~12,500-sample training set (over 12 epochs).

TERM DICTIONARY

Distribution Shift: When subtle patterns in the unseen test data look slightly different from the patterns the model memorized during training.

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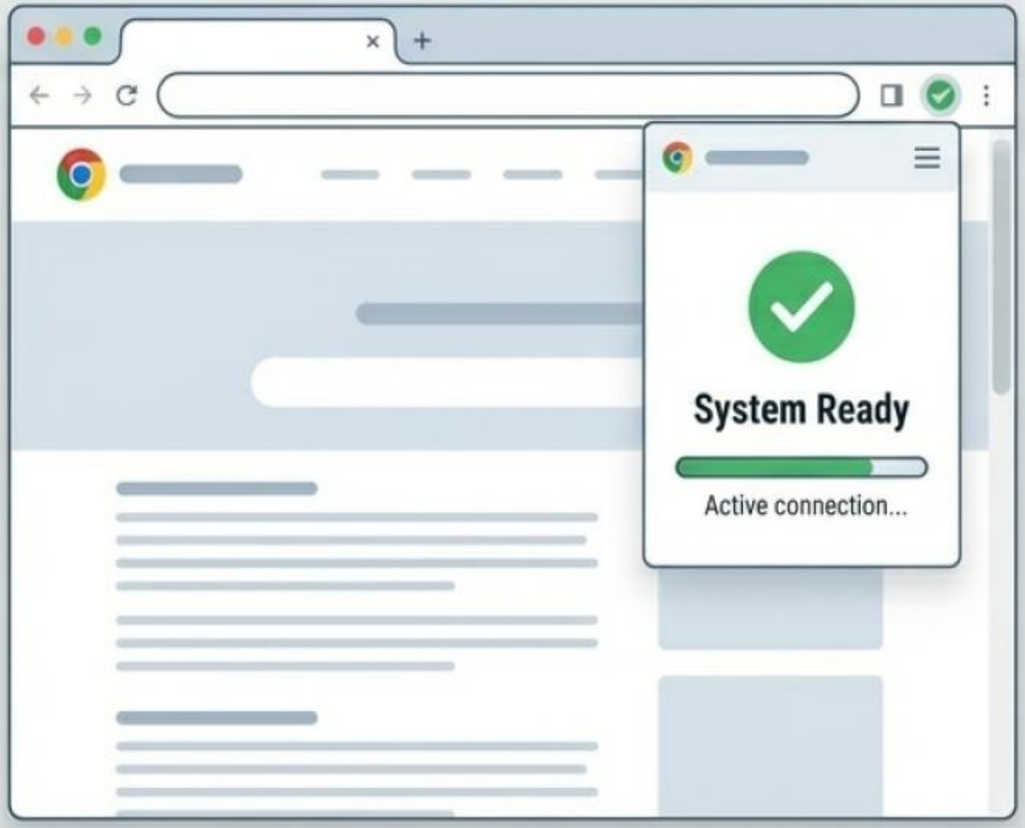
Horizon Phase 1: Web Extension Progress



Web Extension Progress Report

Status Update: The basic Chrome Extension infrastructure is fully operational.

Current Capabilities: Installation on Google Chrome is complete, stable, and capable of reading live page data.

Next Phase: Integration with the LLM (Large Language Model) backend agent is scheduled for the next development phase.

TERM DICTIONARY

Web Extension: A lightweight software program that injects our AI directly into the user's browsing experience.
LLM: An advanced AI model designed to understand and generate complex human language.

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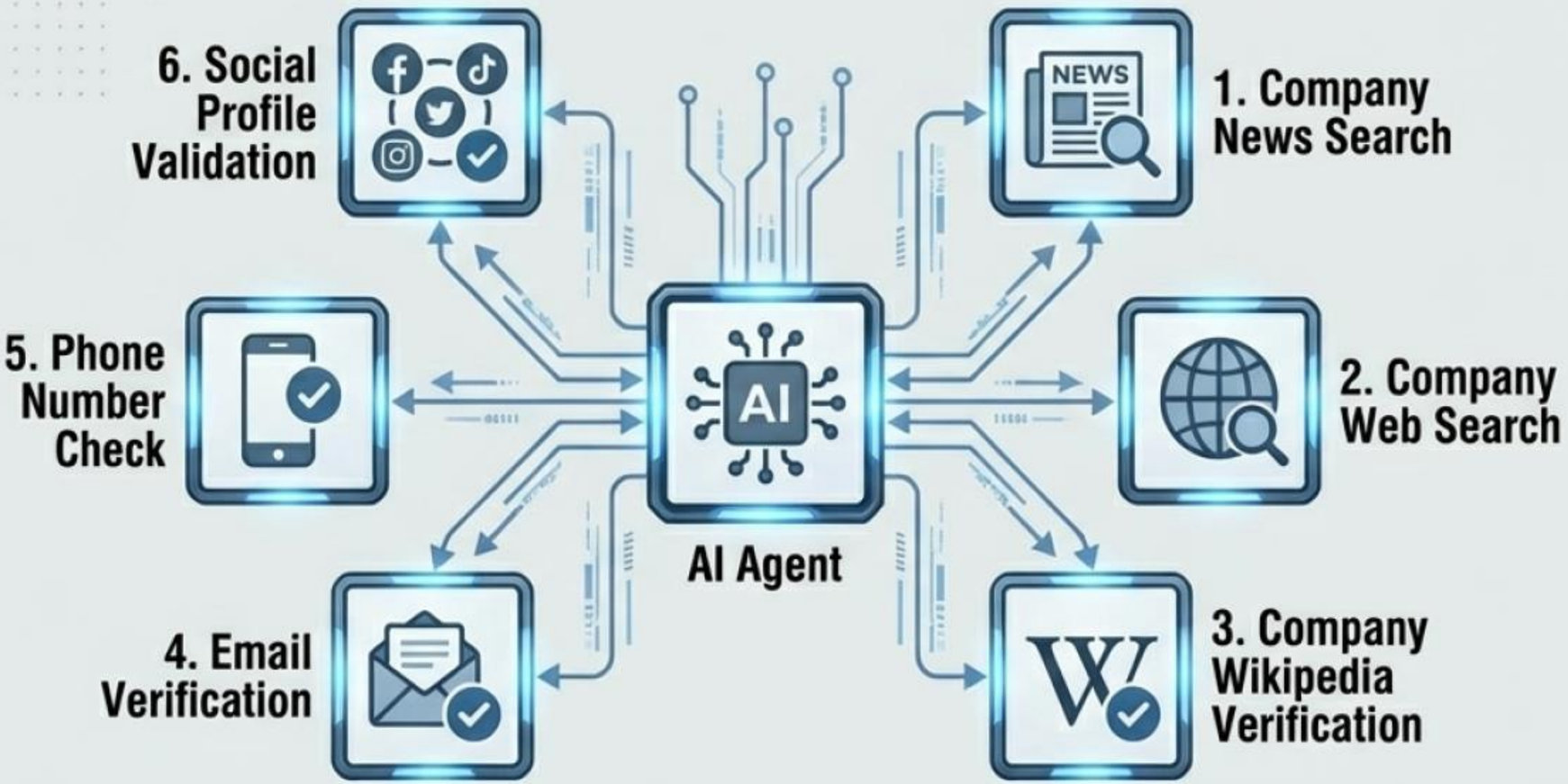
Horizon Phase 2: Agentic Evidence Collection Tools



To move beyond static text analysis, a suite of real-time evidence collection tools has been fully built to cross-reference claims in the wild.

TERM DICTIONARY

Agentic Tools: Independent software scripts that an AI can proactively 'call' to gather live facts from the internet before making a final judgment.

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Team Contribution Tracker: Milestone 5

The final phase was executed through a strongl y collaborative and cross-functional approach, where every team member contributed across planning, development, testing, and validation. Rather than working in isolated roles, the team operated with shared ownership of the entire pipeline—continuously supporting one another in refining models, validating outputs, integrating components, and ensuring overall system reliability. Each member actively participated in discussions, debugging, and iterative improvements, which helped maintain consistency and quality across all deliverables.

While certain members took the lead on specific components—such as documentation, model optimization, integration, or testing—these responsibilities were not siloed. The overlap ensured that knowledge was distributed, decisions were collectively validated, and the final system was both robust and well-understood by the entire team. The milestone's success was ultimately driven by this balanced combination of individual initiative and team-wide collaboration.

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